
UCL COMP0078: Supervised Learning

Problem Sheet

Statistical Learning Theory

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If you find any mistakes/typos, please contact sl-support@cs.ucl.ac.uk.

Overview of Problems

These problems provide justification for some results and proof steps presented in the lecture slides.

- Problem 1: Concentration inequalities to obtain guarantees that hold in high probability rather than only in expectation.
- Problem 2: Understanding how expectation results differ when the function itself depends on data.
- Problem 3: Proof of Popoviciu's inequality providing a bound on the variance of any bounded variable (useful in the lecture slides to bound the generalisation error).
- Problem 4: Quantifying the approximation error on the expected risk using functions thresholded at the p -th decimal.
- Problem 5: Understanding the behaviour of a given bound on the empirical process as the radius of the covering number changes with the sample size.

Problem 1 (slide 9)

Recall that

$$\mathcal{E}_n(f) = \frac{1}{n} \sum_{i=1}^n \ell(f(x_i), y_i) \quad \text{and} \quad \mathcal{E}(f) = \int_{\mathcal{X} \times \mathcal{Y}} \ell(f(x), y) d\rho(x, y) = \mathbb{E}[\mathcal{E}_n(f)].$$

On slide 7, we derive the following bound in expectation

$$\mathbb{E}[|\mathcal{E}_n(f) - \mathcal{E}(f)|] \leq \sqrt{\frac{V_f}{n}}$$

where $V_f = \text{var}(\ell(f(x), y))$.

Can we similarly get a 'tail' bound of the form

$$\mathbb{P}(|\mathcal{E}_n(f) - \mathcal{E}(f)| \leq \varepsilon(n, \delta)) \geq 1 - \delta$$

for some $\varepsilon(n, \delta)$ and $\delta \in (0, 1)$?

Solution 1

We can use Chebyshev's inequality which states that for a random variable X with finite variance and mean, we have

$$\mathbb{P}(|X - \mathbb{E}[X]| > \varepsilon) \leq \frac{\text{var}(X)}{\varepsilon^2}$$

for any $\varepsilon > 0$, or equivalently

$$\mathbb{P}(|X - \mathbb{E}[X]| \leq \varepsilon) \geq 1 - \frac{\text{var}(X)}{\varepsilon^2}.$$

Applying this to our setting with $X = \mathcal{E}_n(f)$, $\mathbb{E}[X] = \mathcal{E}(f)$ and $\text{var}(X) = V_f/n$ where $V_f = \text{var}(\ell(f(x), y))$, we get

$$\mathbb{P}(|\mathcal{E}_n(f) - \mathcal{E}(f)| \leq \varepsilon) \geq 1 - \frac{V_f}{n\varepsilon^2}.$$

Setting $\delta = \frac{V_f}{n\varepsilon^2}$, which gives $\varepsilon = \sqrt{\frac{V_f}{n\delta}}$, we get

$$\mathbb{P}\left(|\mathcal{E}_n(f) - \mathcal{E}(f)| \leq \sqrt{\frac{V_f}{n\delta}}\right) \geq 1 - \delta.$$

This result holds for any $\delta > 0$.

Another possibility is to use Hoeffding's inequality which states that for i.i.d. random variables $X_1, \dots, X_n$, with X_i taking values in $[a_i, b_i]$ for $i \in [n]$, we have

$$\mathbb{P}\left(\left|\sum_{i=1}^n X_i - \sum_{i=1}^n \mathbb{E}[X_i]\right| > t\right) \leq 2 \exp\left(-\frac{2t^2}{\sum_{i=1}^n (b_i - a_i)^2}\right)$$

or equivalently

$$\mathbb{P}\left(\left|\frac{1}{n} \sum_{i=1}^n X_i - \frac{1}{n} \sum_{i=1}^n \mathbb{E}[X_i]\right| > \frac{t}{n}\right) \leq 2 \exp\left(-\left(\frac{t}{n}\right)^2 \frac{2n^2}{\sum_{i=1}^n (b_i - a_i)^2}\right).$$

Setting $\varepsilon = t/n$, this gives

$$\mathbb{P}\left(\left|\frac{1}{n} \sum_{i=1}^n X_i - \frac{1}{n} \sum_{i=1}^n \mathbb{E}[X_i]\right| > \varepsilon\right) \leq 2 \exp\left(-\frac{2n^2\varepsilon^2}{\sum_{i=1}^n (b_i - a_i)^2}\right)$$

or equivalently

$$\mathbb{P}\left(\left|\frac{1}{n} \sum_{i=1}^n X_i - \frac{1}{n} \sum_{i=1}^n \mathbb{E}[X_i]\right| \leq \varepsilon\right) \geq 1 - 2 \exp\left(-\frac{2n^2\varepsilon^2}{\sum_{i=1}^n (b_i - a_i)^2}\right).$$

Applying this to our setting with $X_i = \ell(f(x_i), y_i)$ so that $\frac{1}{n} \sum_{i=1}^n X_i = \mathcal{E}_n(f)$ and $\frac{1}{n} \sum_{i=1}^n \mathbb{E}[X_i] = \mathcal{E}(f)$. Further assuming that the loss ℓ is positive and upper bounded by $C > 0$, we obtain

$$\mathbb{P}(|\mathcal{E}_n(f) - \mathcal{E}(f)| \leq \varepsilon) \geq 1 - 2 \exp\left(-\frac{2n^2\varepsilon^2}{nC^2}\right).$$

Setting $\delta = 2 \exp\left(-\frac{2n\varepsilon^2}{C^2}\right)$, which gives $\varepsilon = C\sqrt{\frac{1}{2n} \log\left(\frac{2}{\delta}\right)}$, we get

$$\mathbb{P}\left(|\mathcal{E}_n(f) - \mathcal{E}(f)| \leq C\sqrt{\frac{1}{2n} \log\left(\frac{2}{\delta}\right)}\right) \geq 1 - \delta.$$

Problem 2 (slide 12)

Explain why

$$\mathbb{E}[\mathcal{E}_n(f)] = \mathcal{E}(f)$$

detailing and justifying every step.

Let $f_n = \underset{f \in \mathcal{F}}{\operatorname{argmin}} \mathcal{E}_n(f)$. Explain why in general

$$\mathbb{E}[\mathcal{E}_n(f_n)] \neq \mathcal{E}(f_n)$$

highlighting the difference with your previous reasoning.

Solution 2

We have

$$\begin{aligned} \mathbb{E}[\mathcal{E}_n(f)] &\stackrel{(i)}{=} \mathbb{E} \left[\frac{1}{n} \sum_{i=1}^n \ell(f(x_i), y_i) \right] \\ &\stackrel{(ii)}{=} \frac{1}{n} \sum_{i=1}^n \mathbb{E} [\ell(f(x_i), y_i)] \\ &\stackrel{(iii)}{=} \frac{1}{n} \sum_{i=1}^n \mathbb{E} [\ell(f(x), y)] \\ &\stackrel{(iv)}{=} \mathbb{E} [\ell(f(x), y)] \\ &\stackrel{(v)}{=} \int_{\mathcal{X} \times \mathcal{Y}} \ell(f(x), y) d\rho(x, y) \\ &\stackrel{(vi)}{=} \mathcal{E}(f) \end{aligned}$$

where

- (i) holds by definition of $\mathcal{E}_n(f)$,
- (ii) holds by linearity of expectation,
- (iii) holds as $(x_i, y_i)_{i=1}^n$ are drawn i.i.d. from ρ which means $\ell(f(x_i), y_i)_{i=1}^n$ are i.i.d.,
- (iv) holds by algebra,
- (v) holds by definition of expectation taken with respect to $(x, y) \sim \rho$, and
- (vi) holds by definition of $\mathcal{E}(f)$.

Now, consider the quantity $\mathbb{E}[\mathcal{E}_n(f_n)]$ where $f_n = \underset{f \in \mathcal{F}}{\operatorname{argmin}} \mathcal{E}_n(f)$. We consider the validity of the above steps with f replaced by f_n . Steps (i) and (ii) still hold. However, step (iii) no longer holds because the terms $\ell(f_n(x_i), y_i)_{i=1}^n$ are not independent: all these terms depend on each other as the function f_n depends on all of $(x_i, y_i)_{i=1}^n$.

The quantity $\mathbb{E}[\mathcal{E}(f_n) - \mathcal{E}_n(f_n)]$ is non-zero and is called the *generalisation error*.

Problem 3 (slide 18)

Prove Popoviciu's inequality: suppose X takes values in $[m, M]$ for some real $m < M$, then

$$\text{var}(X) \leq \frac{(M - m)^2}{4}.$$

Hint: Start by finding the minimum of the function $f(t) = \mathbb{E}[(X - t)^2]$, then compare this minimum to the value $f((M + m)/2)$. You may assume derivatives and expectations can be exchanged without justifications.

Solution 3

Consider the function $f(t) = \mathbb{E}[(X - t)^2]$ for $t \in \mathbb{R}$. It attains its minimum at

$$0 = f'(t) = \frac{d}{dt} \mathbb{E}[(X - t)^2] = \mathbb{E} \left[\frac{d}{dt} ((t - X)^2) \right] = \mathbb{E}[2(t - X)] = 2(t - \mathbb{E}[X])$$

giving $t = \mathbb{E}[X]$. We can double check this is a minimum by noting $f''(t) = 2 > 0$. By properties of the minimum we have

$$f(u) \geq f(\mathbb{E}[X]) = \mathbb{E}[(X - \mathbb{E}[X])^2] = \text{var}(X)$$

for any $u \in \mathbb{R}$. Following the hint, we set $u = (M + m)/2$ to get

$$\begin{aligned} \text{var}(X) &\leq f\left(\frac{M + m}{2}\right) \\ &= \mathbb{E} \left[\left(X - \frac{M + m}{2} \right)^2 \right] \\ &= \frac{1}{4} \mathbb{E} [(2X - M - m)^2] \\ &= \frac{1}{4} \mathbb{E} [(X - m) - (M - X)]^2 \\ &\stackrel{(i)}{=} \frac{1}{4} \mathbb{E} \left[\left| |X - m| - |M - X| \right|^2 \right] \\ &= \frac{1}{4} \mathbb{E} \left[\left| |X - m| - |X - M| \right|^2 \right] \\ &\stackrel{(ii)}{\leq} \frac{1}{4} \mathbb{E} \left[\left| (X - m) - (X - M) \right|^2 \right] \\ &= \frac{1}{4} \mathbb{E} [|M - m|^2] \\ &= \frac{(M - m)^2}{4} \end{aligned}$$

where in step (i) we used the facts that $X - m \geq 0$ and $M - X \geq 0$, and in step (ii) we used the reverse triangle inequality which states that $||x| - |y|| \leq |x - y|$ for all $x, y \in \mathbb{R}$.

Problem 4 (slide 30)

Let $\mathcal{X} = [-1, 1]$, $\mathcal{Y} = \{0, 1\}$ and let ℓ be the misclassification loss. Consider a distribution ρ on $\mathcal{X} \times \mathcal{Y}$ with marginal on $\mathcal{X}$ admitting a density $\rho_{\mathcal{X}}$ which is bounded, that is, $\|\rho_{\mathcal{X}}\|_{\infty} = \sup_{x \in [-1, 1]} \rho_{\mathcal{X}}(x) < \infty$. Recall that the expected risk is $\mathcal{E}(f) = \int_{\mathcal{X} \times \mathcal{Y}} \ell(f(x), y) \rho(x, y) \, dx dy$.

Let $f_* = \operatorname{argmin}_{f \in \mathcal{H}} \mathcal{E}(f)$ be the function at which the global minimum in

$$\mathcal{H} = \{\mathbf{1}_{[a, \infty)} : a \in [-1, 1]\}$$

is attained. Consider the space of functions

$$\mathcal{H}_p = \{\mathbf{1}_{[a, \infty)} : a \in [-1, 1], a10^p = [a10^p]\},$$

in other words the value of a stops at precision 10^{-p} (all real numbers in $[-1, 1]$ truncated after the p -th decimal). Note that $\mathcal{H}_p \subset \mathcal{H}$. Let $f_p = \operatorname{argmin}_{f \in \mathcal{H}_p} \mathcal{E}(f)$. Prove that

$$\mathcal{E}(f_p) - \mathcal{E}(f_*) \leq \|\rho_{\mathcal{X}}\|_{\infty} 10^{-p}.$$

Solution 4

Denote by a_* the element such that $f_* = \mathbf{1}_{[a_*, \infty)}$. Denote by a_p the closest element to a_* in $\mathcal{H}_p$, and let $g_p = \mathbf{1}_{[a_p, \infty)}$. By properties of $\mathcal{H}_p$, we have

$$|a_* - a_p| \leq 10^{-p},$$

and, by definition of f_p , we have

$$\mathcal{E}(f_p) \leq \mathcal{E}(g_p) = \mathcal{E}(\mathbf{1}_{[a_p, \infty)}).$$

Note that $g_p(x) = \mathbf{1}_{[a_p, \infty)}(x)$ and $f_*(x) = \mathbf{1}_{[a_*, \infty)}(x)$ only differ between x is between a_p and a_* . We then have

$$\begin{aligned} \mathcal{E}(f_p) - \mathcal{E}(f_*) &\leq \mathcal{E}(g_p) - \mathcal{E}(f_*) \\ &= \int_{[-1, 1] \times \{0, 1\}} (\ell(g_p(x), y) - \ell(f_*(x), y)) \rho(x, y) \, dy dx \\ &= \int_{-1}^1 \left(\sum_{y \in \{0, 1\}} (\ell(g_p(x), y) - \ell(f_*(x), y)) \rho_{\mathcal{Y}|\mathcal{X}}(y|x) \, dy \right) \rho_{\mathcal{X}}(x) \, dx \\ &\leq \left| \int_{a_p}^{a_*} \left(\sum_{y \in \{0, 1\}} |\ell(g_p(x), y) - \ell(f_*(x), y)| \rho_{\mathcal{Y}|\mathcal{X}}(y|x) \, dy \right) \rho_{\mathcal{X}}(x) \, dx \right| \\ &\leq \left| \int_{a_p}^{a_*} \left(\sum_{y \in \{0, 1\}} 1 \rho_{\mathcal{Y}|\mathcal{X}}(y|x) \, dy \right) \rho_{\mathcal{X}}(x) \, dx \right| \\ &= \left| \int_{a_p}^{a_*} \rho_{\mathcal{X}}(x) \, dx \right| \\ &\leq \|\rho_{\mathcal{X}}\|_{\infty} |a_* - a_p| \\ &\leq \|\rho_{\mathcal{X}}\|_{\infty} 10^{-p}, \end{aligned}$$

where we have used the fact that $|\ell(g_p(x), y) - \ell(f_*(x), y)|$ is at most 1 is the misclassification loss only takes value either 0 or 1. Note that the absolute value around the integral is only needed to allow for the possibility that $a_* < a_p$.

Problem 5 (slide 46)

Recall that

$$\mathbb{E} \left[\sup_{f \in \mathcal{H}} |\mathcal{E}(f_n) - \mathcal{E}(f)| \right] \leq 2L\eta + \mathcal{N}(\mathcal{H}, \eta) \sqrt{\frac{V_{\mathcal{H}}}{n}}$$

where $\mathcal{N}(\mathcal{H}, \eta)$ is the covering number of radius $\eta > 0$, and $V_{\mathcal{H}} = \sup_{f \in \mathcal{H}} \text{var}(\ell(f(x), y))$.

The covering number of a ball of radius R in $\mathbb{R}^d$ is upper bounded by $(4R/\eta)^d$. Setting $\eta = n^r$ for some $r \in \mathbb{R}$, how should r be chosen in order to guarantee the above upper bound tends to 0 as n tends to infinity?

Solution 5

First, when $\mathcal{H}$ is a ball of radius R in $\mathbb{R}^d$, the bound becomes

$$\mathbb{E} \left[\sup_{f \in \mathcal{H}} |\mathcal{E}(f_n) - \mathcal{E}(f)| \right] \leq 2L\eta + \left(\frac{4R}{\eta} \right)^d \sqrt{\frac{V_{\mathcal{H}}}{n}} = \mathcal{O}(\eta + \eta^{-d} n^{-1/2})$$

In order for the upper bound to tend to 0, we need both terms η and $\eta^{-d} n^{-1/2}$ to tend to 0. Letting $\eta = n^r$ for some $r \in \mathbb{R}$, the condition $\eta \rightarrow 0$ gives $r < 0$. The second condition is $\eta^{-d} n^{-1/2} = n^{-rd-1/2} \rightarrow 0$ which requires $-rd - 1/2 < 0$ or equivalently $r > -\frac{1}{2d}$.

We deduce that the upper bound tends to zero when $\eta = n^r$ for $-\frac{1}{2d} < r < 0$ which tends to zero but extremely slowly (especially when the dimension d is large). That is, η needs to converge to zero but at an even slower rate than $n^{-1/(2d)}$.